Gong Cheng

Academic Position

since 2026 **Associate Professor**, *College of Surveying and Geo-informatics, Tongji University, Shanghai*, China.

2023–2026 **Research Scientist, Lecture**, *Department of Earth Sciences, Dartmouth College, Hanover, NH*, USA.

2021–2023 **Postdoctoral Researcher**, *Department of Earth Sciences, Dartmouth College, Hanover, NH*, USA.

2020–2021 **Postdoctoral Researcher**, *Department of Earth System Science, University of California, Irvine, CA*, USA.

Education

2014–2019 **Ph.D in Computational Science**, *Uppsala University*, Sweden.
Thesis: Numerical ice sheet modeling-Forward and inverse problems. Dissertation on 18 October 2019.
Thesis available at: http://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-392268

2012–2014 **Master of Computational Science**, *Uppsala University*, Sweden.

2009–2012 **Master of Biomedical Engineering**, *Zhejiang University*, Hangzhou, China.

2005–2009 **Bachelor of Biomedical Engineering**, *Zhejiang University*, Hangzhou, China.

2005–2007 **Mixed Honors Class**, *Zhejiang University*, Hangzhou, China.
Mixed Honors Class is an honors program for the top 5% students of Zhejiang University. 139 students are selected out of approximately 3500 freshmen majoring in science and engineering at the whole university, based on their high school GPA, performance in screening tests and personal interviews.

Grants

2026-2029 **NSF 25-530 Collaborations in Artificial Intelligence and Geosciences**, *Collaborative Research: CAIG: Physics-informed Machine Learning of Ice Calving and Sliding*, (PI, $505,036).

2026-2027 **Astera Institute**, *A Model-Ready Living Data Product Built with Physics-Informed Neural Networks for Greenland & Antarctica*, (PI, $93,574 without overhead).

2024-2027 **NASA Research Opportunities in Space and Earth Sciences (ROSES) - Cryospheric Science**, *Reconstructing the evolution of Helheim Glacier between 1981 and 2023 by combining numerical modeling and remote sensing data*, (PI, $354,812).

Publications

26* **ISSM-ML: a modular framework for integrating machine learning models into the Ice-sheet and Sea-level System Model (ISSM v2026.4)**, *Y. Liu, M. Morlighem, G. Cheng*, Geoscientific Model Development, https://doi.org/10.5194/egusphere-2026-6193 under review, 2026.

25* **Frontal and Basal Processes Jointly Shape Seasonal Ice Flow of Tidewater Glaciers in Western Greenland**, *K. Oniszk, J. Badgeley, G. Cheng, W. Colgan, S. Khan*, Communications Earth and Environment, under review, 2026.

24 **The implementation and effectiveness of Calving Algorithms in numerical ice models (CalvingMIP)**, *J. R. Jordan, F. Pattyn, D. Abele, T. Albrecht, J. Alvarez-Solas, J. M. Barnes, T. Berends, J. A. Bernales, J. Blasco, G. Cheng, Y. Choi, S. L. Cornford, C. Garcia-Molina, F. Gillet-Chaulet, G. H. Gudmundsson, A. Humbert, G. R. Leguy, W. H. Lipscomb, M. Montoya, D. Moreno-Parada, M. Morlighem, T. Pelle, A. Robinson, M. Rückamp, H. Seroussi, Y. Tian, L. Wagner, R. S. W. van de Wal, L. Zhao, and T. Zwinger*, The Cryosphere Discussion, accepted, 2026.

23 **Standardizing Glacier Centerline Starting-Point Delineation Using Velocity Thresholds: A Length-Sensitivity Analysis of Antarctic Peninsula Glaciers**, *G. Zheng, Y. Wu, G. Cheng, L. An, R. Li*, Advances in Polar Science, Accepted, 2026.

22 **A vast valley network beneath the Greenland Ice Sheet**, *A. Chartrand, J. MacGregor, M. Morlighem, O. Sergienko, H. Ockenden, W. Colgan, G. Cheng*, Geophysical Research Letters, https://doi.org/10.1029/2026GL122028 53 (2026), e2026GL122028.

21 **Inferring subglacial topography using physics informed machine learning constrained by two conservation laws**, *M. Krishna, G. Cheng and M. Morlighem*, The Cryosphere, https://doi.org/10.5194/tc-20-3533-2026 20 (2026): 3533–3558.

20 **Future Retreat of Great Aletsch Glacier and Hintereisferner – application of a full-Stokes model to two valley glaciers in the European Alps**, *M. Rückamp, G. Cheng, K. Gutjahr, M. Möller, P. K.E. Pellikka, and C. Mayer*, The Cryosphere, https://doi.org/10.5194/tc-20-2999-2026 20 (2026): 2999–3024.

19 **DJ4Earth: Differentiable, and Performance-portable Earth System Modeling via Program Transformations**, *W. Moses, G. Cheng, V. Churavy, M. Gelbrecht, M. Klöwer, J. Kump, M. Morlighem, S. Williamson, D. Apte, P. Berg, M. Giordano, C. Hill, N. Loose, A. Montoison, S. Narayanan, A. Pal, M. Schanen, S. Silvestri, G. Wagner, P. Heimbach*, Journal of Advances in Modeling Earth System, https://doi.org/10.1029/2025MS005615 , 18, e2025MS005615.

18 **DeepTopoNet: A Framework for Subglacial Topography Estimation on the Greenland Ice Sheets**, *B. Tama, M. Krishna, H. Alam, M. Cham, O. Faruque, G. Cheng, J. Wang, M. Morlighem, V. Janeja*, Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems, Association for Computing Machinery, New York, NY, USA, https://doi.org/10.1145/3748636.3764162 , 943–954, 2025.

17 **A Python library for solving ice sheet modeling problems using Physics Informed Neural Networks, PINNICLE v1.0**, *G. Cheng, M. Krishna and M. Morlighem*, Geoscientific Model Development, https://doi.org/10.5194/gmd-18-5311-2025 , 18 (2025): 5311–5327.

16 **Calving Parameterization of Helheim Glacier, East Greenland, using Graph Neural Network Emulators for Numerical Ice Sheet Models**, *Y. Koo, G. Cheng, M. Morlighem and M. Rahnemoonfar*, The Cryosphere, https://doi.org/10.5194/tc-19-2583-2025 , 19 (2025): 2583–2599.

15 **Ice front and Mélange Dynamics at Kangerlussuaq Glacier, eastern Greenland, since 1981, and future stability**, *E. Lippert, M. Morlighem, G. Cheng, D. Berg, J. Hassan and S. Khan*, Geophysical Research Letters, https://doi.org/10.1029/2024GL113982 52 (2025), e2024GL113982.

14 **Monthly Greenland ice sheet elevation changes during 2003-2023 reveal complex evolving mass change pattern**, *S. Khan, H. Seroussi, M. Morlighem, W. Colgan, V. Helm, G. Cheng, D. Berg, V. R. Barletta, N. K. Larsen, W. Kochtitzky, M. van den Broeke, K. H. Kjær, A. Aschwanden, B. Noël, J. E. Box, J. A. MacGregor, R. S. Fausto, K. D. Mankoff, I. M. Howat, K. Oniszk, D. Fahrner, A. Lokkegaard, E. Y. H. Lippert, A. Bratner, and J. Hassan*, Earth System Science Data, https://doi.org/10.5194/essd-17-3047-2025 17 (2025): 3047–3071.

13 **Ice thickness-induced variations in effective pressure and basal conditions influence seasonal and multi-annual ice velocity at Sermeq Kujalleq (Jakobshavn Isbræ)**, *X. Lu, A. Sole, S. Livingstone, G. Cheng, L. Jiang, T. Chudley, B. Noël and D. Li*, Geophysical Research Letters, https://doi.org/10.1029/2024GL111092 52 (2025): e2024GL111092.

12 **Forward and Inverse Modeling of Ice Sheet Flow Using Physics-Informed Neural Networks: Application to Helheim Glacier, Greenland**, *G. Cheng, M. Morlighem, and S. Francis* , JGR: Machine Learning and Computation, https://doi.org/10.1029/2024JH000169 1.3 (2024), e2024JH000169.

11 **Numerical stabilization methods for level-set-based ice front migration**, *G. Cheng, M. Morlighem and H. Gudmundsson*, Geoscientific Model Development, (2024) https://doi.org/10.5194/gmd-17-6227-2024 .

10 **Modeling a century of change: Kangerlussuaq Glacier's mass loss from 1933 to 2021**, *E. Lippert, M. Morlighem, G. Cheng and S. Khan*, Geophysical Research Letters, https://doi.org/10.1029/2023GL106286 51.4 (2024): e2023GL106286.

9 **Evaluation of four calving laws for Antarctic ice shelves**, *J. Wilner, M. Morlighem and G. Cheng*, The Cryosphere, https://doi.org/10.5194/tc-17-4889-2023 17 (2023): 4889–4901.

8 **Helheim Glacier's Terminus Position Controls Its Seasonal and Inter-Annual Ice Flow Variability**, *G. Cheng, M. Morlighem, J. Mouginot, and D. Cheng*, Geophysical Research Letters, https://doi.org/10.1029/2021GL097085 49.5 (2022): e2021GL097085.

7 **Sensitivity of ice sheet surface velocity and elevation to variations in basal friction and topography in the Full Stokes and Shallow Shelf Approximation frameworks**, *G. Cheng, N. Kirchner and P. Lötstedt*, The Cryosphere, https://doi.org/10.5194/tc-15-715-2021 15.2 (2021): 715-742.

6 **Parameter sensitivity analysis of dynamic ice sheet models-Numerical computations**, *G. Cheng and P. Lötstedt*, The Cryosphere, https://doi.org/10.5194/tc-14-673-2020 14.2 (2020): 673-691.

5 **A full Stokes subgrid scheme in two dimensions for simulation of grounding line migration in ice sheets using Elmer/ICE (v8.3)**, *G. Cheng, P. Lötstedt and L. von Sydow*, Geoscientific Model Development, https://doi.org/10.5194/gmd-13-2245-2020 13 (2020): 2245–2258.

4 **Thermal conductivity of firn at Lomonosovfonna, Svalbard, derived from subsurface temperature measurements**, *S. Marchenko, G. Cheng, P. Lötstedt, V. Pohjola, R. Pettersson, W. van Pelt, and C. Reijmer*, The Cryosphere, https://doi.org/10.5194/tc-13-1843-2019 13.7 (2019): 1843-1859.

3 **Dynamically coupling Full Stokes and Shallow Shelf Approximation for marine ice sheet flow using Elmer/Ice (v8. 3)**, *E. CH van Dongen, N. Kirchner, M. B. van Gijzen, R. SW van de Wal, T. Zwinger, G. Cheng, P. Lötstedt, and L. von Sydow*, Geoscientific Model Development, https://doi.org/10.5194/gmd-11-4563-2018 11 (2018): 4563-4576.

2 **Anisotropic Radial Basis Function Methods for Continental Size Ice Sheet Simulations**, *G. Cheng and V. Shcherbakov*, Journal of Computational Physics, https://doi.org/10.1016/j.jcp.2018.06.020 372 (2018): 161-177.

1 **Accurate and stable time stepping in ice sheet modeling**, *G. Cheng, P. Lötstedt, and L. von Sydow*, Journal of Computational Physics, https://doi.org/10.1016/j.jcp.2016.10.060 329 (2017): 29-47.

Supervising Experiences

Since 2025 Co-supervise Ph.D. student, Sina Marie Felten, Reconstruction of Jakobshavn Isbræ from 1850s, DTU, Denmark

since 2025 Supervise Postdoc, Yinmin Liu, Reconstructing The Evolution Of Helheim Glacier Between 1981 And 2023 By Combining Numerical Modeling And Remote Sensing Data, Dartmouth College, USA

since 2023 Co-supervise Ph.D. student, Mansa Krishna, Inferring sub-glacial bed topography with physics-informed machine learning, Dartmouth College, USA

2023-2024 Supervise Bachelor research project, Artificial Intelligence Ice Flow Models for Sea Level Rise Projections: leveraging Physics-Informed Neural Networks (PINNs) to predict ice sheet behavior in a changing climate, Sade Francis, Dartmouth College, USA

2022-2024 Co-supervise Ph.D. student, Eigil Lippert, Numerical glacier flow modelling in Greenland, DTU, Denmark

2022-2023 Co-supervise Ph.D. student, Joel Wilner, Evaluating and Comparing Calving Laws in Antarctica, Dartmouth College, USA

2019 Supervise Master thesis: Numerical ice sheet modeling using FEniCS, André Löfgren, Uppsala University, Sweden

2018 Supervise student project: Ice Sheet Modeling: Accuracy of First-Order Stokes Model with Basal Sliding, Eskil Jonsson, Uppsala University, Sweden

2017 Supervise Master thesis: Coupling approximation levels for modeling ice flow on paleo time scales, Eef van Dongen, Uppsala University

2016 Supervise Bachelor student project: Observational Sensitivity to Sliding Coefficient in Shallow Ice Approximation, Anton Sundin, Uppsala University

2016 Supervise Bachelor student project: Quantification of the effects of perturbed bedrock data on the surface of moving ice sheets, Fredrik Lauren, Joakim Johansson, Uppsala University

Teaching Experiences

2025 EARS 70/170: Glaciology, Dartmouth College, USA

2022 GEOG 15.01: Climate Change, Dartmouth College, USA

2021 ESS 17: Catastrophes, UC Irvine, USA

2019 Programming in Python(Master level, 5 ECTS), Uppsala University, Sweden

2019 Scientific Computing, Bridging Course(Master level, 5 ECTS), Uppsala University, Sweden

2018 Applied Finite Element Methods(Master level, 5 ECTS), Uppsala University, Sweden

2014-2017 Optimization(Master level, 5 ECTS), teaching assistant, Uppsala University, Sweden

Attended Conferences & Talks

2026 **International Symposium on Artificial Intelligence in Glaciology**, *Hanover, NH, USA*, June 2026, Oral presentation, A Physics-Informed Neural Network Framework for Forward and Inverse Ice-Sheet Modeling.

2026 **EGU General Assembly 2026**, *Vienna, Austria*, May 2026, Poster, When the Ground and the Glacier Disagree: The Timing Mystery of Bedrock Uplift and Ice Discharge Peaks.

2025 **AGU Fall meeting 2025**, *New Orleans, LA, USA*, December 2025, Oral presentation, Identifying drivers of seasonal mass loss for Greenland's three largest outlet glaciers using numerical modeling and GNSS Observations.

2025 **Artificial Intelligence and Digital Twins for Earth Systems**, *Austin, TX, USA*, September 2025, Oral presentation, A Python library for solving ice sheet modeling problems using Physics Informed Neural Networks, PINNICLE.

2025 **BACO 2025**, *Busan, Republic of Korea*, July 2025, Oral presentation, PINNICLE: Physics-Informed Neural Networks for Ice and CLimatE.

2025 **The Future of Earth's Polar Regions**, *Santa Barbara, CA, USA*, June 2025, Invited talk, A Python Library for Solving Ice Sheet Modeling Problems Using a Unified Framework.

2025 **EGU General Assembly 2025**, *Vienna, Austria*, May 2025, Oral presentation, Identifying the Basal Friction Law Using Numerical Models and GNSS Observations.

2025 **SIAM Conference on Computational Science and Engineering (CSE25)**, *Fort-Worth, TX, USA*, Mar 2025, Oral presentation, A Python Library for Solving Ice Sheet Modeling Problems Using a Unified Framework.

2024 **AGU Fall meeting 2024**, *Washington D.C., USA*, December 2024, Oral presentation, Physics-Informed Neural Networks for Ice and CLimatE (PINNICLE).

2024 **SIAM Conference on Mathematics of Planet Earth (MPE24)**, *Portland, OR, USA*, June 2024, Mini Tutorial, Differentiable Earth System Models in Julia.

2024 **EGU General Assembly 2024**, *Vienna, Austria*, April 2024, Poster, A Unified Framework for Forward and Inverse Modeling of Ice Sheet Flow using Physics Informed Neural Network.

2024 **Future of Greenland ice Sheet Science Workshop 2024**, *Moscow, ID, USA*, April 2024, Oral presentation, Neural networks for forward and inverse numerical modeling.

2023 **AGU Fall meeting 2023**, *San Francisco, CA, USA*, December 2023, Oral presentation, Using deep learning to constrain poorly represented basal processes in ice sheet models.

2023 **Mathematics on Ice Forum**, *online*, May 2023, Invited talk, Helheim Glacier's terminus position controls its seasonal and inter-annual ice flow variability.

2022 **AGU Fall meeting 2022**, *Chicago, IL, USA*, December 2022, Poster, Investigating the drivers of Helheim Glacier's variability from 2007 to 2020.

2022 **Seminar series in numerical analysis, KTH & Stockholm University**, *Stockholm, Sweden*, April 2022, Invited talk, Parameter sensitivity analysis of dynamic ice sheet models.

2021 **AGU Fall meeting 2021**, *New Orleans, LA, USA*, December 2021, Poster, Helheim Glacier's terminus position controls its seasonal and inter-annual ice flow variability.

2021 **TDB seminar at IT department, Uppsala University**, *Uppsala, Sweden*, October 2021, Invited talk, Investigating the drivers of Helheim Glacier's variability using numerical modeling.

2021 **SIAM Conference on Mathematical and Computational Issues in the Geosciences**, *online*, June 2021, Oral presentation, Sensitivity Study of Helheim Glacier using Automatic Differentiation.

2021 **EGU General Assembly 2021**, *online*, April 2021, Oral presentation, Investigating the drivers of Helheim Glacier's variability from 2007 to 2020.

2020 **AGU Fall meeting 2020**, *online*, December 2020, Oral presentation, Investigating the drivers of Helheim Glacier's variability from 2007 to 2020.

2019 **SIAM Conference on Mathematical and Computational Issues in the Geosciences**, *Houston, USA*, March 2019, Oral presentation, Efficient Numerical Ice-sheet Simulations over Long Time Spans.

2018 **International Glaciological Society - Nordic Branch Meeting**, *Rovaniemi, Finland*, October 2018, Oral presentation, Sub-grid modeling of grounding line migration problem.

2018 **13th World Congress in Computational Mechanics**, *New York City, USA*, July 2018, Oral presentation, Efficient Numerical Ice-Sheet Simulations over Long Time Spans.

2017 **International Glaciological Society - Nordic Branch Meeting**, *Uppsala, Sweden*, October 2017, Oral presentation, A meshfree approach to ice sheet simulations.

2017 **SIAM Conference on Mathematical and Computational Issues in the Geosciences**, *Erlangen, Germany*, September 2017, Oral presentation, Efficient Numerical Ice-Sheet Simulations over Long Time Spans.

2017 **SIAM Annual Meeting (AN17)**, *Pittsburgh, USA*, July 2017, Oral presentation, Efficient Numerical Ice-Sheet Simulations over Long Time Spans.

2017 **SIAM Conference on Computational Science and Engineering(CSE17)**, *Atlanta, USA*, February 2017, Oral presentation, Accurate and Stable Time Stepping in Ice Sheet Modeling.

2015 **International Glaciological Society - Nordic Branch Meeting**, *Copenhagen, Denmark*, October 2015, Poster, Accurate and Stable Time Stepping in Ice Sheet Modeling.

2015 **Karthaus Summer School**, *Karthaus, Italy*, September 2015.

Professional Activities

since 2024 Editor for the EGU's journal: The Cryosphere

since 2024 Founder and core developer of the Physics Informed Neural Networks for Ice and CLimatE (PINNICLE)

since 2022 Core developer of the differentiable JUlia ICE model (DJUICE.jl)

since 2020 Core developer of the Ice-sheet and Sea-level System model (ISSM)

2026 Instructor of The China Land Ice Modeling Bootcamp (CLIMB)

2026 Instructor of The Glaciology and Machine Learning Summer School (GlaMacLeS)

2026, 2025 Convener for AGU Session: Machine Learning in the Cryosphere

2027, 2026 Convener for EGU Session: Modelling ice sheets and glaciers

since 2021 Member of the European Geosciences Union

since 2020 Member of the American Geophysical Union

2017–2018 President of Uppsala University Student SIAM Chapter, Sweden

since 2015 Member of the Society for Industrial and Applied Mathematics

Journal review Nature Communications, Geophysical Research Letters, The Cryosphere, Geoscientific Model Development, Journal of Geophysical Research: Earth Surface, Journal of Glaciology, Advances in Climate Change Research

Proposal review National Science Foundation, The Research Council of Norway, NASA FINESST, Netherlands Organisation for Scientific Research